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29 August 1969
SUPERSEDING
MIL-S-2047
4 December 1951

MILITARY SPECIFICATION

SIGNAL, GROUND, YELLOW SMOKE, M64
PARTS FOR AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the parts and the loading, assembling and packing of one signal designated as Signal, Ground, Yellow Smoke, M64 (For use with Grenade Launchers M7 and M8).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

C-F-206 - Felt, Sheet, Wood, Pressed
QQ-P-416 - Plating, Cadmium (Electrodeposited)

MILITARY

MIL-P-223 - Powder, Black
JAN-P-663 - Powder, Fuze
MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

FEDERAL

FED-STD-151 - Metals; Test Methods

FSC: 1370

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-286 - Propellants; Sampling, Inspection and Testing
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1234 - Pyrotechnics: Sampling, Inspection and Testing
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

8866684 - Container, Ammunition, Fiber M104A1 for Ground Signals
8866685 - Box, Packing, Ammunition for Ground Signals in Fiber Containers M104A1
9235593 - Signal, Ground, Yellow Smoke, M64 (For use with Grenade Launchers M7 and M8) Assembly

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET - 9235593 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.- Materials and parts shall be in accordance with applicable drawings and specifications.

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3.2 Assembly.- The assembly shall comply with all requirements specified on Drawing (dwg.) 9235593, with all requirements of all drawings referenced thereon, and with all requirements specified in applicable specifications.

3.3 Moisture content

3.3.1 Moisture content of propelling charge.- The moisture content of the propelling charge at the time of loading shall not exceed 0.3 percent when tested as specified in 4.4.1.4.

3.4 Waterproofness.- The signal shall show no evidence of leakage when subjected to test specified in 4.4.2.

3.5 Functioning.- The signal shall function and comply with the following requirements when tested as specified in 4.4.3.

3.5.1 Premature burst.- The assembly shall not burst at an altitude of less than 150 feet, and the smoke assembly shall not eject within 100 feet of the ground.

3.5.2 The signal shall not fail to emit the required color smoke (see dwg. 9235591).

3.5.3 Ejection and burning time.- The smoke assembly shall not fail to eject and burn for the required time (see dwg. 9235591).

3.5.4 Altitude.- No smoke assembly shall emit the first trace of smoke at an altitude of less than 600 feet and the average altitude for all smoke assemblies which function above 600 feet shall be not less than 700 feet.

3.6 Workmanship.- All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined and all packing components shall be dry.

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3.7 First article inspection.- This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.- At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Statement of findings (certificates of analysis) on all compositions and powders, and certificates of conformance on all remaining materials purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.
h. A statement that the components and assemblies were dried, ventilated and assembled as required by the applicable drawings.

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1. A statement that the assemblies were assembled in areas remote from solvents and volatiles as required by the drawings.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.- The contractor shall submit a first article sample as designated by the contracting officer (see 6.1c) for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of the following quantities which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract:

a. Twenty-five (25) complete sets of inert parts (i.e. 25 each of every inert component and sub-assembly).

b. Thirty-two (32) loaded signal assemblies completely packaged and sealed as required by the applicable drawings.

All parts and materials, including packaging and packing components, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and he shall submit a record of this inspection with the sample, including statements of findings and certificates of conformance of materials (see 4.1.1e). A sample containing known defects shall not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government. Unless otherwise directed by the contracting officer, loaded components shall not be submitted for any test until the inert components first article has been accepted.

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4.2.2 First article inspection provisions.

4.2.2.1 Inert parts

4.2.2.1.1 Submission.- The 25 sets of inert parts (see 4.2.1a) shall be submitted to a Government approved laboratory designated by the contracting officer.

4.2.2.1.2 Inspections to be performed.- The inert parts will be subjected to any or all of the applicable examinations and tests specified in 4.3.2 and 4.3.3 of this specification and any or all of the requirements of the applicable drawings.

4.2.2.1.3 Rejection.- If any inert component or sub-assembly fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an inert component or sub-assembly to comply with any of the stated requirements.

4.2.2.2 Loaded signals

4.2.2.2.1 Submission.- The 32 loaded signals (see 4.2.1b) shall be submitted to a Government Proving Ground designated by the contracting officer.

4.2.2.2.2 Inspections to be performed.- The loaded signals will be subjected to the functioning test (see 3.5) in accordance with the method specified in 4.4.3.

4.2.2.2.3 Rejection.- The first article sample shall be rejected if any loaded signal fails to comply with any or all of the requirements of 3.5. The Government reserves the right to terminate its inspection upon encountering any failure of an item in the sample to comply with any or all requirements.

4.3 Inspection provisions

4.3.1 Lot formation.- The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample

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selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105. Unless otherwise approved by the contracting officer the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots of components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot of signal assemblies shall contain:

a. Parts of one kind from one supplier.

b. Black powder from not more than one lot from one supplier.

c. Composition and smoke charge ingredients from not more than one lot from one supplier.

d. Fuze composition from one batch or blending of batches that results in homogeneity of the composition.

e. Smoke charge composition produced by one manufacturer under one continuous set of operating conditions and which consists of one or more batches that have been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

4.3.2 Examination.- Inspection for Critical defects (and Major defects, when so specified) shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual

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characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Housing, fuze (see dwg. 9207854 covering a detail of dwg. 9235593).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Pitch diameter of thread, minimum (min.)	Gage	01001
102.	Major diameter of thread, min.	Gage	01002
103.	Evidence of poor workmanship in charge groove and charge hole (see 3.6)	Visual	01003
104.	Intermediate charge hole obstructed	Visual	01004
Minor:	AQL 6.50 percent		
201.	Pitch diameter of thread, maximum (max.)	Gage	01005
202.	Diameter of body surface, min.	Gage	01006
203.	Diameter of flange	Gage	01007
204.	Thickness of flange	Gage	01008
205.	Diameter of body crimping groove	Gage	01009
206.	Length of thread pilot	Gage	01010
207.	Length from threaded end to flange	Gage	01011
208.	Length from body end to flange	Gage	01012
209.	Inside diameter before ring groove, max	Gage	01013
210.	Location of intermediate charge hole	Gage	01014
211.	Location of igniting charge cavity	Gage	01015

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212.	Large diameter of fuze charge groove, max.....	Gage	01016
213.	Small diameter of fuze charge groove, min.....	Gage	01017
214.	Depth of powder pellet cavity...	Gage	01018
215.	Depth of inner charge cavity....	Gage	01019
216.	Diameter of inner charge cavity.	Gage	01020
217.	Location of inner charge cavity.	Gage	01021
218.	Diameter of powder pellet cavity.....	Gage	01022
219.	Diameter of intermediate charge cavity.....	Gage	01023
220.	Depth of seal ring and fuze charge cavity.....	Gage	01024
221.	Length of thread, min.....	Gage	01025
222.	Location of body crimping groove.....	Gage	01026
223.	Diameter of igniting charge cavity.....	Gage	01027
224.	Diameter of igniting charge intersecting cavity.....	Gage	01028
225.	Depth of inside diameter before ring groove.....	Gage	01029
226.	Finish improper.....	Visual	01030
227.	Evidence of poor workmanship except in charge groove or charge hole (see 3.6).....	Visual	01031

4.3.2.2 Body (see dwg. 9235587 covering a detail of dwg. 9235593).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Crack or split.....	Visual	02001
Minor:	AQL 2.50 percent		
201.	Inside diameter, max.....	Gage	02002
202.	Outside diameter, min.....	Gage	02003

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203.	Length.....	Gage	02004
204.	Ends not square.....	Visual	02005
205.	Evidence of poor workmanship (see 3.6).....	Visual	02006

4.3.2.3 Body assembly (see dwg. 9235588 covering a detail of dwg. 9235593).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Any charge missing.....	Visual	03001
102.	Ring seal flush or above.....	Visual	03002
Minor:	AQL 2.50 percent		
201.	Seal missing between body and fuze housing.....	Visual	03003
202.	Body insecure.....	Manual	03004
203.	Crimp missing or size incorrect.	Visual	03005
204.	Retaining disc insecure.....	Manual	03006
205.	Evidence of poor workmanship (see 3.6).....	Visual	03007

4.3.2.4 Fin (see dwg. 8848616 covering a detail of dwg. 9235593).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Outside diameter.....	Gage	04001
102.	Thickness, min.....	Gage	04002
103.	Crack or fracture.....	Visual	04003
104.	Weld missing, mislocated or of incorrect size.....	Visual	04004

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Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 2.50 percent		
201.	Width of formed section.....	Gage	04005
202.	Width of ring either side of formed section.....	Gage	04006
203.	Location of formed section (circular).....	Gage	04007
204.	Inside diameter, max.....	Gage	04008
205.	Length of overlap.....	Gage	04009
206.	Total length, min.....	Gage	04010
207.	Evidence of poor workmanship (see 3.6).....	Visual	04011

4.3.2.5 Tube, stabilizer (see dwg. 8848615 covering a detail of dwg. 9235593).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Crack or fracture.....	Visual	05001
Major: None defined			
Minor:	AQL 4.00 percent		
201.	Pitch diameter of thread, max...	Gage	05002
202.	Minor diameter of thread, max...	Gage	05003
203.	Total length.....	Gage	05004
204.	Inside diameter, max.....	Gage	05005
205.	Wall thickness forward of rear end, min.....	Gage	05006
206.	Wall thickness below thread, min.....	Gage	05007
207.	Length to end of threads, min...	Gage	05008
208.	Depth of thread counterbore, max.....	Gage	05009
209.	Length of tapered sidewall, max.	Gage	05010
210.	Depth of threaded cavity.....	Gage	05011
211.	Outside diameter, min.....	Gage	05012
212.	Evidence of poor workmanship (see 3.6).....	Visual	05013

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4.3.2.6 Stabilizer assembly (see dwg. 8848618 covering a detail of dwg. 9235593).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Inside diameter, min.....	Gage	06001
102.	Threads damaged.....	Visual	06002
103.	Fin missing.....	Visual	06003
Minor:	AQL 1.50 percent		
201.	Length, max.....	Gage	06004
202.	Location of fin on tube over maximum.....	Gage	06005
203.	Protective coating missing or contains bare spot.....	Visual	06006
204.	Evidence of poor workmanship (see 3.6).....	Visual	06007

4.3.2.7 Top, identification (see dwg. 9235592 covering a detail of dwg. 9235593).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Crack or split.....	Visual	07001
Minor:	AQL 2.50 percent		
201.	Outside diameter.....	Gage	07002
202.	Small diameter of mating surface	Gage	07003
203.	Large diameter of mating surface	Gage	07004
204.	Length.....	Gage	07005
205.	Protective coating missing or contains bare spot.....	Visual	07006
206.	Embossing missing.....	Visual	07007
207.	Evidence of poor workmanship (see 3.6).....	Visual	07008

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4.3.2.8 Cup (see dwg. 9235586 covering a detail of dwg. 9235593).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Inside diameter, max.....	Gage	08001
102.	Length, min.....	Gage	08002
Minor:	AQL 1.00 percent		
201.	Diameter of hole.....	Gage	08003
202.	Evidence of poor workmanship (see 3.6).....	Visual	08004

4.3.2.9 Smoke pellet assembly, prior to inserting smoke composition (see dwg. 9235591 covering a detail of dwg. 9235593).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Priming charge missing.....	Visual	09001
Minor: None defined			

4.3.2.10 Smoke pellet assembly after inserting smoke composition (see dwg. 9235591 covering a detail of dwg. 9235593).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Smoke composition missing.....	Visual	10001
Minor: None defined			

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4.3.2.11 Smoke pellet assembly prior to crimping (see dwg. 9235591 covering a detail of dwg. 9235593).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Priming charge missing.....	Visual	11001
Minor: None defined			

4.3.2.12 Smoke pellet assembly (see dwg. 9235591 covering a detail of dwg. 9235593).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Loose or leaking charge.....	Visual	12001
Major:	AQL 0.65 percent		
101.	Hole surface not completely covered with ignition composition.....	Visual	12002
102.	Crimped metal not pressed firmly against disc for 360 degrees....	Visual	12003
103.	Cup ruptured.....	Visual	12004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6).....	Visual	12005

4.3.2.13 Assembly, prior to assembling identification top, propelling charge or stabilizer assembly (see dwg. 9235593).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			

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Major:	AQL 0.40 percent		
101.	Any component missing or damaged to extent that function may be impaired.....	Visual	13001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6).....	Visual	13002

4.3.2.14 Assembly, prior to assembling disc, disc lining, and stabilizer assembly (see dwg. 9235593).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Weight of propelling charge.....	Balance	14001
102.	Propelling charge missing.....	Visual	14002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6).....	Visual	14003

4.3.2.15 Assembly, prior to assembling stabilizer assembly (see dwg. 9235593).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Identification top missing, damaged, insecure, not completely seated or unsealed...	Visual-Manual	15002
102.	Disc incorrectly applied or ruptured to extent that charge is exposed or leaking.....	Visual	15003

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Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6).....	Visual	15004

4.3.2.16 Assembly, prior to the drying of the pettman cement (see dwg. 9235593).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Cement missing from stabilizer and fuze housing joint.....	Visual	16001
Minor:	None defined		

4.3.2.17 Assembly (see dwg. 9235593).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Coating color incorrect.....	Visual	17001
2.	Designation incorrect.....	Visual	17002
Major:	AQL 0.65 percent		
101.	Misalignment, max.....	Gage	17003
102.	Assembly damaged to extent that function may be impaired.....	Visual	17004
103.	Stabilizer assembly insecure....	Visual-Manual	17005
104.	Plug missing or insecure.....	Visual-Manual	17006
Minor:	AQL 2.50 percent		
201.	Total length, max.....	Gage	17007
202.	Diameter at fin, max.....	Gage	17008
203.	Marking misleading or unidentifiable.....	Visual	17009

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204.	Plug tape missing, insecurely tied, twisted or waterproofing omitted.....	Visual-Manual	17010
205.	Evidence of poor workmanship (see 3.6).....	Visual	17011

4.3.2.18 Sealed fiber container (see dwg. 8866684).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Container cut or damaged through all impregnated layers.....	Visual	18001
102.	Sealing strip incomplete or badly wrinkled.....	Visual	18002
103.	Metal end loose or distorted....	Visual	18003
Minor:	AQL 2.50 percent		
201.	Gap between body and cover of container in excess of 1/8 inch.	Visual-Manual	18004
202.	Contents loose.....	Manual	18005
203.	Tear tab length improper.....	Visual	18006
204.	Color of sealing strip incorrect.....	Visual	18007
205.	Marking misleading or unidentifiable.....	Visual	18008

4.3.2.19 Sealed wood packing box (see dwg. 8866685).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.65 percent		
101.	Box damaged to extent that contents are exposed or liable to become exposed.....	Visual	19001
102.	Hardware broken or loose.....	Visual-Manual	19002
103.	Strapping broken or loose.....	Visual-Manual	19003
104.	DOD symbol missing or unidentifiable.....	Visual	19004
Minor:	AQL 2.50 percent		
201.	Hardware improperly engaged.....	Visual-Manual	19005
202.	Strapping improperly assembled..	Visual-Manual	19006
203.	Car seal missing, unsealed or mislocated.....	Visual-Manual	19007
204.	Contents loose.....	Manual	19008
205.	Marking misleading or unidentifiable.....	Visual	19009

4.3.3 Production testing

4.3.3.1 Moisture content (see Table I) - Major defect

TABLE I

Material	Code No.
Black powder (see dwg. 9235636)	20001
Fuze powder (see dwg. 9235636)	20002
Smoke composition (see dwg. 9235591)	20003
Cushioning wad (see dwg. 9235593)	20004
Propelling charge (see 3.3.1)	20005

Moisture content at the time of loading (see Table I).- The contractor shall provide adequate controls to insure compliance with the requirements, and shall test for verification at least one sample of each powder, composition and material used in each eight (8) hours loading production of signal assemblies. A composite

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sample shall not be used. If the moisture content of a sample exceeds the requirement, that quantity or sub-lot of powder, composition and material represented by the sample shall be rejected. If powder, composition and material from a sub-lot with excessive moisture has been used in loading, all assemblies containing the non-conforming ingredients shall be rejected. The tests shall be performed as specified in 4.4.1.

4.3.3.2 Waterproofness (see 3.4) - Major defect - Code No. 21001.- Thirty-two (32) signal assemblies shall be selected from each lot for test. If any sample fails to comply with the requirement, the lot shall be rejected. The test shall be performed as specified in 4.4.2.

4.3.3.3 Functioning (see 3.5 and 6.5)

4.3.3.3.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, 125 samples shall be selected from each lot for test. The lot shall be rejected if during the test, a critical defect occurs, the average altitude is less than required, or if eight (8) or more assemblies exhibit any of the remaining major defects listed in Table II. The test shall be performed as specified in 4.4.3.

TABLE II

Defect	Classification	Code No.
Premature burst (see 3.5.1)	Critical	17012
Incorrect color (see dwg. 9235591)	Critical	17013
Burning time incorrect (see dwg. 9235591)	Major	17014
Altitude less than min. (see 3.5.4)	Major	17015
Average altitude less than min. (see 3.5.4)	Major	17016
Signal fails to function (propelling charge fails to initiate) (see 3.5)	Major	17017
Smoke charge fails to eject (see 3.5.3)	Major	17018

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4.3.3.3.2 After three consecutive lots have complied with the criteria of 4.3.3.3.1, fifty (50) samples shall be selected from each lot for test. The lot shall be rejected if during the test, a critical defect occurs, the average altitude is less than required, or if four (4) or more assemblies exhibit any of the remaining major defects listed in Table II.

4.3.3.4 Hardness of stabilizer tube (see dwg. 8848615) - Major defect - Code No. 05014.- Thirty-two (32) stabilizer tubes shall be selected from each heat treated batch for test. If any sample fails to comply with the requirement, the batch represented by the sample shall be rejected. The test shall be performed as specified in 4.4.4.

4.3.3.5 Load test of stabilizer welds (see dwg. 8848618) - Major defect - Code No. 06008.- Thirty-two (32) stabilizer assemblies shall be selected from each lot for test. If any sample fails to comply with the requirements, the lot shall be rejected. The test shall be performed as specified in 4.4.5.

4.3.3.6 Protective coating test of stabilizer assembly (see dwg. 8848618) - Code No. 06009 and identification top (see dwg. 9235592) - Code No. 07009 - Major defects.- Sampling, acceptance and rejection shall be in accordance with QQ-P-416, except that actual parts may be used in lieu of specimens. The test procedure shall be as specified in 4.4.6.

4.3.4 Inspection equipment.- Equipment Tabulation Number ET - 9235593 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

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4.3.4.2 Supply and maintenance.- Supply and maintenance of the equipment listed on the ET shall be in accordance with MIL-I-45607.

4.4 Test methods and procedures

4.4.1 Moisture content

4.4.1.1 Black powder.- The moisture content of the black powder shall be determined as specified in MIL-P-223 using a sample weight of approximately 0.5 gram and heating thirty (30) minutes at 70 degrees Centigrade (C.).

4.4.1.2 Fuze powder.- The moisture content of the fuze powder shall be determined as specified in JAN-P-663.

4.4.1.3 Smoke composition.- The moisture content of the smoke composition shall be determined in accordance with the method specified in MIL-STD-1234, Method 101.1, except that the test drying time shall be four (4) hours and vacuum shall be applied.

4.4.1.4 Propelling charge.- The moisture content of the propelling charge shall be determined as specified in MIL-STD-286, Method 102.1.

4.4.1.5 Cushioning wad.- The moisture content of the wad shall be determined in accordance with C-F-206.

4.4.2 Waterproofness.- The sample signals shall be immersed horizontally in water at 70 plus or minus 10 degrees Fahrenheit (F.) to a depth of not less then 6 inches, nor more then 7 inches, over the highest part for 15 minutes. Remove from the water and wipe the exterior surfaces dry. Within one hour after removal, disassemble the signal and examine the interior of body and fuze housing and all component parts for evidence of water.

4.4.3 Functioning.- This test shall be performed at a Government Proving Ground. The signals shall be fired from a Grenade Launcher, M7, without retaining spring, mounted on an M1 Rifle, with the rifle in a vertical position and the butt of the rifle resting on normal solid ground, or on a suitable mounting which will give approximately the same altitude of functioning.

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Signals shall not be fired when wind velocity exceeds 16 miles per hour. The assembly shall be observed for compliance with the requirements.

4.4.4 Hardness of the stabilizer tube.- The stabilizer tube shall be tested as specified in FED-STD-151, Method 243.

4.4.5 Load test of stabilizer welds.- The load test of the stabilizer welds shall be performed in accordance with the procedure and equipment specified in 4.3.4.

4.4.6 Protective coating of stabilizer assembly and identification top.- The adequacy of the protective coating shall be tested in accordance with the procedures in QQ-P-416.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.- The signal assemblies shall be preserved and packaged in accordance with dwg. 8866684.

5.2 Packing

5.2.1 Level A.- The sealed containers shall be packed in accordance with dwg. 8866685.

5.3 Marking.- Marking of the containers and packing boxes shall be in accordance with dwgs. 8866684 and 8866685.

6. NOTES

6.1 Ordering data.- Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.- Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Submission of first article quantity.- Instructions as to the location of the facility at which the evaluation of the first article quantity shall be carried out may be obtained from the contracting officer.

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6.2 Inspection code numbers.- The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.- The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).- Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.- The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.- Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared

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designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.- Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.- All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND15. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

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6.4 Proving Ground test summary

Lot	Sample Size	Test	Requirements
First Article	32	Functioning	See Table II
Regular Production (First three consecutive acceptable lots)	125	Functioning	See Table II
Remaining	50	Functioning	See Tabel II

6.4.1 Test Validity.-If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

6.5 Initial Production Testing.-If the provisions of AMCR 700-34, which requires Initial Production Testing, apply, such testing shall be accomplished in accordance with USATECOM Project No. 8-MU-008-064-001.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1370-A-280

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